

# Classical Machine Learning Model Selection

## A Practical Guide to Non-Neural-Network Algorithms for Plant and Process Engineers

Industrial AI Capability Building Workshop

### 1 The Algorithms at a Glance

| Method & Task                                                | How It Works                                                                                                                   | Strengths                                                                                                | Watch-Outs                                                                                                                                                    |
|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Linear Regression</b><br>[Regression]                     | Fits a straight-line (planar) relationship $\hat{y} = \mathbf{w}^\top \mathbf{x} + b$ by least squares.                        | Fast and transparent: each coefficient is the effect of one input. The natural first baseline.           | Misses non-linear behaviour; sensitive to outliers; coefficients become unstable when inputs are strongly correlated.                                         |
| <b>Ridge Regression</b><br>[Regression]                      | Linear regression with a penalty on large coefficients ( $L_2$ : $\lambda \sum w_j^2$ ) that shrinks them towards zero.        | Stays stable when sensors are correlated (e.g. neighbouring thermocouples).                              | Keeps every input (no variable selection); still sensitive to outliers.                                                                                       |
| <b>Lasso Regression</b><br>[Regression]                      | Linear regression with an $L_1$ penalty ( $\lambda \sum  w_j $ ) that can set coefficients exactly to zero.                    | Automatic variable selection, giving short, simple models.                                               | From a group of correlated inputs, keeps one somewhat arbitrarily and drops the rest.                                                                         |
| <b>Elastic Net</b><br>[Regression]                           | Combines the Ridge and Lasso penalties.                                                                                        | Selects variables while coping with groups of correlated inputs.                                         | Two settings to tune: penalty strength and the Lasso/Ridge mix.                                                                                               |
| <b>Logistic Regression</b><br>[Classification]               | A linear model passed through an S-shaped curve, $p = 1/(1 + e^{-z})$ , to give the probability of a class.                    | Outputs probabilities (e.g. chance a batch fails QC); effects read as odds ratios; very fast.            | Straight-line boundary between classes; non-linear effects must be added as engineered features.                                                              |
| <b>SVM / SVR</b><br>[Regression & Classification]            | Finds the boundary (or, for regression, a tolerance band) with the widest margin; kernels allow curved relationships.          | Works well on small-to-medium datasets with smooth non-linear behaviour.                                 | Slow on large datasets; inputs must be scaled; several settings to tune.                                                                                      |
| <b>Decision Tree</b><br>[Regression & Classification]        | A sequence of yes/no splits on input values (e.g. "Temp > 72 °C?") ending in a prediction.                                     | Readable rules; captures thresholds; no need to scale inputs.                                            | Unstable (small data changes give a different tree); overfits unless depth is limited; step-wise predictions.                                                 |
| <b>Random Forest</b><br>[Regression & Classification]        | Averages hundreds of trees, each trained on a random resample of the data, with a random subset of inputs tried at each split. | Good accuracy with little tuning; handles non-linearity and interactions; tolerant of noisy data.        | Larger and slower than a single tree; built-in importance scores favour inputs with many distinct values.                                                     |
| <b>XGBoost / LightGBM</b><br>[Regression & Classification]   | Gradient boosting: builds trees one after another, each correcting the errors of the ones before.                              | Usually the most accurate choice for tabular plant data; handles missing values; fast.                   | Needs tuning; can overfit small or noisy datasets; sensitive to outliers in the target.                                                                       |
| <b>k-Nearest Neighbours</b><br>[Regression & Classification] | Predicts from the $k$ most similar past records (average or majority vote).                                                    | Simple, no training step; easy to explain ("these five past batches looked like this").                  | Slow predictions on large datasets; inputs must be scaled; degrades with many inputs.                                                                         |
| <b>Isolation Forest</b><br>[Anomaly Detection]               | Random splits separate unusual points quickly; the fewer splits needed to isolate a point, the more anomalous it is.           | Needs no labelled faults; fast on large datasets; good for spikes and unusual combinations of readings.  | Judges each row on its own, so add lag or rolling features for time-based anomalies; the alarm threshold must be set; weaker when anomalies cluster together. |
| <b>PCA</b><br>[Unsupervised]                                 | Re-expresses many correlated variables as a few uncorrelated components that capture most of the variation.                    | Compresses sensor arrays; basis of multivariate process monitoring (Hotelling's $T^2$ and SPE/Q charts). | Linear only; components are blends of variables (inspect the loadings); sensitive to outliers; standardise inputs first.                                      |

#### Before You Pick an Algorithm: Four Rules That Matter More

- **Features beat algorithms.** Physically meaningful inputs (deviation from setpoint, rates of change, ratios, time since cleaning) usually improve results more than switching models.
- **Split by time, not at random.** Train on earlier data and test on later data; random splits leak future information and overstate accuracy.
- **Rare events need the right score.** If only 2% of batches fail, a model that always predicts "pass" is 98% accurate. Judge fault and defect models by precision and recall.
- **Scale inputs where needed.** Standardise inputs (mean 0, standard deviation 1) for Ridge/Lasso, SVM,  $k$ -NN and PCA; tree-based models do not need it.

## 2 Decision Framework: Which Model to Choose?

### Three-Step Model Selection

#### 1. Define the target

- **[Regression]** A number (reject rate %, fill volume, temperature) → **XGBoost/LightGBM, Random Forest** or **Ridge/Elastic Net**.
- **[Classification]** A category (fault code, QC pass/fail) → **XGBoost/LightGBM, Random Forest** or **Logistic Regression**.
- **[Anomaly Detection]** No labels; looking for abnormal behaviour → **Isolation Forest** (individual abnormal readings) or **PCA with  $T^2$ /SPE charts** (multivariate drift).

#### 2. Match the model to the process and the data

- *Behaviour is roughly linear, or the model must be fully auditable:*
  - A few inputs, not strongly correlated → **Linear** or **Logistic Regression**.
  - Strongly correlated sensors (e.g. temperature/pressure arrays) → **Ridge**.
  - Many candidate inputs, most probably irrelevant → **Lasso** or **Elastic Net**.
- *Thresholds, interactions or step changes are present:*
  - Typical plant dataset (roughly 1,000 rows or more) → **XGBoost/LightGBM (default choice)**.
  - Noisy data or little time to tune → **Random Forest**.
  - Small dataset (a few hundred rows) with a smooth non-linear response → **SVR**.

#### 3. Explain and validate

- **Linear models:** compare standardised coefficients (reliable only when inputs are not strongly correlated).
- **Tree ensembles:** use **SHAP** (SHapley Additive exPlanations):
  - *Summary (beeswarm) plot:* which inputs matter most, and in which direction.
  - *Dependence plot:* where the model's response changes (e.g. the line speed above which predicted rejects rise).
  - *Waterfall plot:* why the model made one specific prediction (e.g. for a single shift or batch).
- **SHAP explains the model, not the process.** Confirm apparent root causes with process knowledge or plant trials; correlated sensors can share or swap credit.

## 3 Summary Scorecard

| Algorithm             | Task         | Accuracy       | Training Speed     | Ease of Interpretation | Tuning Effort | Outlier Robustness     |
|-----------------------|--------------|----------------|--------------------|------------------------|---------------|------------------------|
| Linear Regression     | Reg.         | Low–Moderate   | Very fast          | Very high              | Minimal       | Low                    |
| Ridge / Lasso / E-Net | Reg.         | Moderate       | Very fast          | High                   | Low           | Low                    |
| Logistic Regression   | Class.       | Low–Moderate   | Very fast          | Very high              | Minimal       | Low–Moderate           |
| Decision Tree         | Both         | Moderate       | Fast               | High (if shallow)      | Low           | High                   |
| Random Forest         | Both         | High           | Moderate           | Moderate (with SHAP)   | Low           | High                   |
| XGBoost / LightGBM    | Both         | Very high      | Fast               | Moderate (with SHAP)   | Moderate–High | Moderate               |
| SVM / SVR             | Both         | Moderate–High  | Slow on large data | Low                    | High          | Moderate               |
| k-Nearest Neighbours  | Both         | Low–Moderate   | None needed        | Moderate               | Low           | Low                    |
| Isolation Forest      | Anomaly      | Data-dependent | Fast               | Low–Moderate           | Low           | N/A (detects outliers) |
| PCA                   | Unsupervised | N/A            | Fast               | Low–Moderate           | Low           | Low                    |

*Accuracy* refers to typical performance on tabular plant data and always depends on the dataset. *Outlier robustness:* tree models tolerate extreme input values; boosting with default settings is sensitive to extreme target values. *Both* = regression and classification.